

Free AI Agent - README

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1 Project Setup and Execution

1.1 Instructions to Set Up and Run the Project

1. Clone the repository:

```
git clone <your-repository-url>  
cd <your-repository-folder>
```

2. Create a virtual environment (recommended):

```
python -m venv venv  
source venv/bin/activate # On Windows use: venv\Scripts\activate
```

3. Install dependencies:

```
pip install -r requirements.txt
```

4. Run the application:

```
streamlit run app.py
```

5. Open the browser and navigate to the provided local URL.

2 Dependencies Required

- Python 3.x
- Streamlit
- Transformers (Hugging Face library)
- Any additional required libraries (see requirements.txt)

3 Working Application

3.1 Frontend and Backend Integration

The frontend, built using Streamlit, sends user queries to the backend. The backend processes these queries using a GPT-2 AI model and returns responses, which are then displayed on the frontend.

4 Documentation

4.1 How the AI Agent Works

- The AI agent is powered by GPT-2, a text-generation model.
- The user inputs a query, and the model generates a response based on conversation history.
- Streamlit provides a simple UI to interact with the AI model.

4.2 Challenges Faced and Solutions

- **Model Performance:** GPT-2 may sometimes produce incomplete or irrelevant responses. Adjusting parameters like `max_length` helps.
- **Latency:** Loading the model initially takes time. Using `@st.cache_resource` improves performance.
- **State Management:** Maintaining conversation history was challenging, but Streamlit's session state resolved this.

5 License

This project is open-source and available under the MIT License.